

PROJECT LANDFALL

The Living World Navigation System

Governing Architecture and Product Requirements Document

Geolocation, Living Charts, Creator Waypoint Authoring, Contextual Precision, and Native Real-World Chronicle Integration



Governing Principle

A Chronicle must be able to guide a Player through the real world using only the Creator tools and the Player device. Physical markers, installed beacons, QR codes, NFC tags, or custom hardware may exist as optional extensions, but they are never prerequisites for the core Landfall experience.

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Prepared for the Chronicles platform

Document Control

Field	Value
Program	Project Landfall
Subsystem	The Living World Navigation System
Document type	Governing architecture and product requirements
Version	1.0
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Status	Governing baseline
Primary product surfaces	Creator Studio, Voyage Chart, Player Journal, Chronicle runtime, native companion app
Core implementation rule	Zero-infrastructure by default; physical-world markers and hardware are optional providers only

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1. Executive Summary

Project Landfall establishes the Chronicles platform as a native real-world navigation and location-aware storytelling system. It allows Creators to place waypoints, zones, routes, and location-triggered story moments on a live map, then lets Players experience those places through a Chronicle-styled Living Chart rather than a generic mapping product pasted into the interface.

The subsystem is designed around a zero-infrastructure default. A Creator can build a complete location-enabled Chronicle without installing signs, placing QR codes, attaching NFC tags, deploying Bluetooth beacons, or arranging special equipment at the destination. Those methods may later exist as optional verification providers, but Landfall must remain useful when the only physical-world resources are the place itself and the Player's phone.

Landfall integrates with the existing Voyage Chart as the canonical Player map surface. It also introduces an embedded Living Chart block and compact navigation cues, all driven by the same authoritative map, waypoint, route, and progression state. There must never be one map system for the Voyage Chart and another unrelated one inside story blocks. That way lies duplicate state, mismatched pins, and the usual software archaeology expedition.

The core system combines browser or native location services, a styleable vector-map renderer, accuracy-aware verification, optional contextual signals, route and zone authoring, privacy-safe storage, and Chronicle progression events. The system reports confidence rather than pretending every blue dot is divine revelation.

Core Product Outcome

A Creator visually designs a real-world journey. A Player follows that journey through a beautifully integrated Chronicle map. The platform knows when the Player is nearby, when the evidence is good enough to confirm arrival, and how to continue gracefully when location is unavailable.

2. Project Identity and Product Vision

2.1 Program Name

The subsystem is named Project Landfall.

Landfall is the moment a journey reaches a real place. The name fits a system that connects the authored Chronicle world to physical geography without implying that every experience must be nautical, outdoors, or long-distance. A museum gallery, a city block, a forest trail, a campus, and a single historic building can all become a Landfall experience.

2.2 Formal Subtitle

The Living World Navigation System

2.3 Product Vision

Landfall should make the real world feel authored without making it feel artificial. The map, the Player marker, the route, the waypoint reveal, the confidence state, and the arrival moment should all use the Chronicle's own language, materials, animation, sound, and visual hierarchy. The Player should feel that the journal understands where the journey is occurring, not that the experience opened a miniature third-party map application in a rectangular iframe.

2.4 Primary Use Cases

- Outdoor treasure hunts across parks, towns, campuses, trails, shorelines, and islands.
- Museum or historic-site Chronicles using floor maps, natural landmarks, room-level guidance, and contextual confirmation.
- Date-night journeys with scenic waypoints, timed transitions, hidden destinations, and location-triggered messages.
- Educational field experiences where arrival unlocks explanations, questions, media, or artifacts.
- City exploration, walking tours, scavenger hunts, and multi-stop narrative routes.
- Indoor or compact experiences that use existing architecture, exhibits, or visual landmarks rather than installed markers.
- Hybrid fictional maps aligned to real geography, allowing an authored parchment chart to guide a physical journey.

2.5 Product Non-Goals

- Landfall is not a turn-by-turn automotive navigation replacement.
- Landfall is not a covert tracking or workforce-monitoring platform.
- Landfall does not promise room-level or exhibit-level precision from ordinary GPS alone.
- Landfall does not require Creators to install infrastructure in the real world.
- Landfall does not treat location as an anti-cheat or security-grade identity proof.
- Landfall does not expose raw Player location history to public community content or ordinary Creator analytics.
- Landfall does not fork the Voyage Chart into a second incompatible map domain.

3. Non-Negotiable Design Principles

Principle	Requirement
Zero-Infrastructure Default	Every core feature must work with only the Creator Studio and the Player device. Physical markers and hardware remain optional.
Chronicle-Native Presentation	The map must inherit Chronicle themes, materials, language, motion, focus, and progression state. It must not look like an unrelated mapping website.
One Living Chart	Voyage Chart, embedded map blocks, compact navigation cues, Captain previews, and replay views are projections of one canonical Landfall state.
Confidence, Not Fiction	The system must represent uncertainty, accuracy, staleness, and confidence honestly. A noisy reading may be “nearby,” not “confirmed.”
Progression Remains Authoritative	Location evidence proposes or satisfies a completion condition. It does not bypass Tale Session event rules, idempotency, visibility, or replay safety.
Privacy by Data Minimization	Store completion evidence by default, not a continuous trail. Raw samples should be processed locally and discarded unless a narrowly defined feature requires retention.
Fair Failure	Permission denial, weak signal, indoor failure, offline operation, and incompatible devices must all provide an understandable fallback path.
Creator Guidance Over Creator Guesswork	Studio must warn when a radius, map style, route, or privacy choice is unrealistic or unsafe.
Provider Independence	Map rendering, map data, geocoding, elevation, and native location services must be replaceable behind explicit interfaces.

Principle	Requirement
Accessible Wayfinding	Location, bearing, proximity, route status, and arrival must never rely only on moving graphics, color, vibration, or sound.

Hard Boundary

A Creator may choose optional QR, NFC, beacon, UWB, or installed-marker providers in later versions, but no core Landfall acceptance gate may depend on them. The system must be excellent before any physical accessory enters the room.

4. Capability Tiers

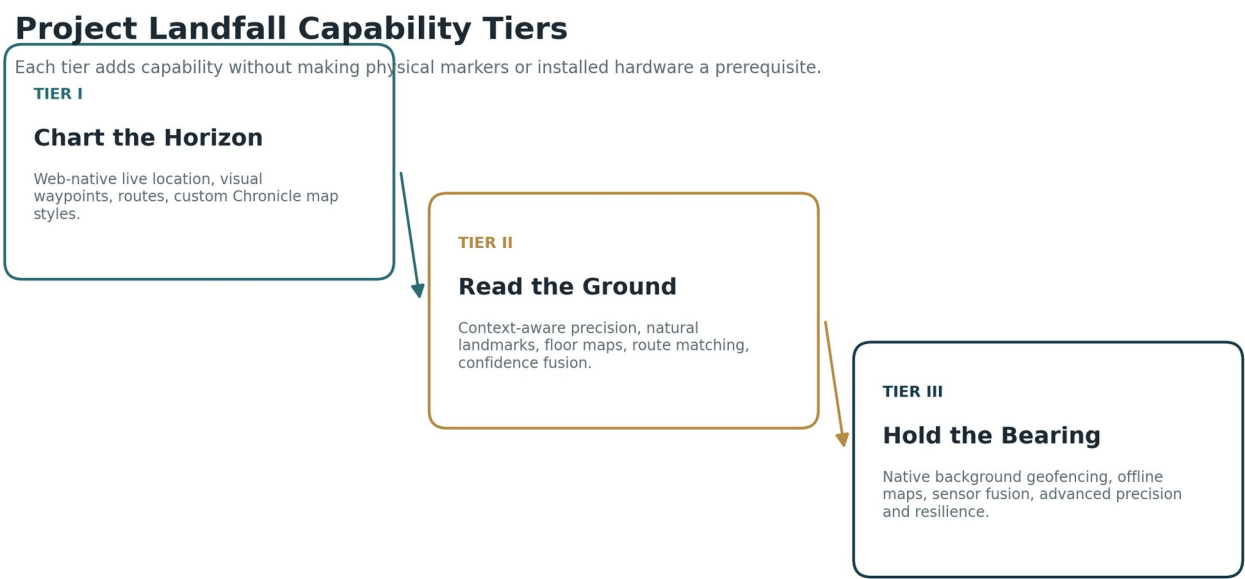


Figure 1. Project Landfall capability tiers.

4.1 Tier I: Chart the Horizon

Tier I delivers the complete web-native foundation and should support most outdoor treasure hunts immediately. It uses the browser Geolocation API in secure contexts, a styleable vector-map renderer, Creator-placed waypoints and routes, accuracy-aware foreground tracking, and Chronicle progression integration. The W3C Geolocation API provides position data, reported accuracy, and one-shot or watched updates; access requires explicit user permission and is commonly restricted to HTTPS. [R1] [R2]

- Visual waypoint and zone placement on a live map.
- Route drawing, ordering, branching, and optional-route behavior.
- Live Player location with an accuracy halo and confidence state.
- Distance, bearing, dwell, arrival, and nearby-state calculation.
- Voyage Chart integration, Living Chart block, and compact navigation cue.
- Multiple map themes and Creator-customized visual treatment.
- Location-gated Story Blocks and location-triggered reveals.
- Foreground tracking while the Player is actively using the relevant Chronicle surface.
- Privacy classification, consent, and data-minimized completion evidence.

- Offline-safe already-downloaded maps and Chronicle content where supported by the web platform.

4.2 Tier II: Read the Ground

Tier II improves precision and usefulness in dense, compact, or indoor environments without requiring the Creator to install anything. It combines ordinary position readings with map constraints, route matching, device motion, heading, elevation hints, floor-plan overlays, Creator-authored natural landmarks, and optional Vision Waypoint recognition of existing scenery or exhibits.

- Creator-uploaded and aligned floor maps or custom site maps.
- Natural landmark reference sets made from existing places, signs, architecture, or exhibits.
- Visual-inertial and motion-assisted continuity while the Player walks through a site.
- Route corridor matching and impossible-jump rejection.
- Floor, room, wing, or zone inference when the available evidence supports it.
- Progressive guidance: property reached, building reached, region reached, target likely nearby.
- Location plus Vision Waypoint confirmation using the real environment, not installed markers.
- Creator test walks that record actual accuracy and confidence behavior before publication.

4.3 Tier III: Hold the Bearing

Tier III uses a native companion application to add resilient background behavior, offline map packages, OS-managed geofences, stronger sensor fusion, and advanced device capabilities. Apple Core Location supports standard and significant location updates, region monitoring, and beacon ranging; Android provides geofencing APIs but recommends broad geofence radii for reliable background triggering. [R3] [R4]

- Background geofence arrival and departure events under OS permission rules.
- Native offline map regions and route packages.
- Location updates that continue through screen lock or route transitions where policy permits.
- Improved integration of heading, inertial motion, barometer, and platform location services.
- Background notification and safe return to the correct Chronicle moment.
- Advanced optional providers such as UWB or beacons, without making them mandatory.
- Native power, privacy, and permission management.

5. Canonical Architecture

Canonical Landfall Architecture

One geospatial domain, several natural views. The Voyage Chart is the canonical Player surface.

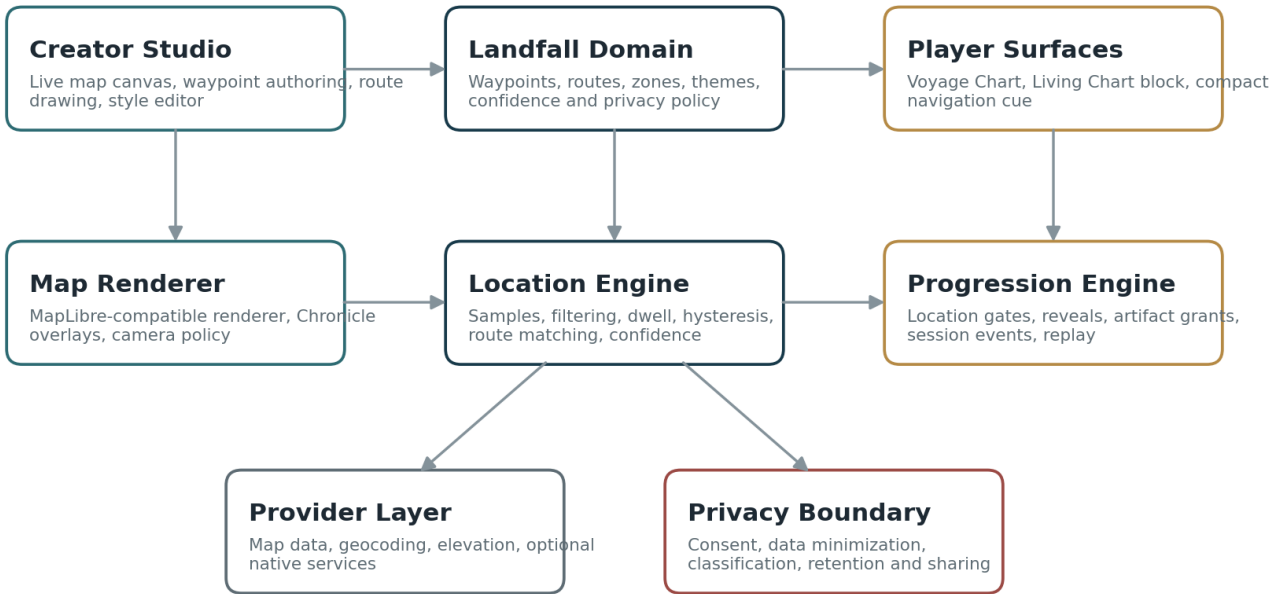


Figure 2. Canonical Landfall architecture and ownership boundaries.

5.1 Aggregate Boundaries

Landfall must be a canonical geospatial subsystem beneath Creator Studio, the Tale publishing pipeline, Tale Sessions, and Player presentation. It should not be implemented as a collection of map components that each fetch location and interpret arrival independently.

Aggregate	Owns	Must not own
Landfall Map Definition	Map identity, data source, style preset, camera policy, custom overlays, attribution metadata	Player progress or current device location
Waypoint Definition	Geometry, radius or region, precision profile, dwell, hints, privacy class, completion policy	Runtime completion state
Route Definition	Ordered or branching geometry, corridors, sequence rules, directionality, travel mode	Tale Session mutation
Landfall Runtime State	Current location confidence, active waypoint, nearby state, route progress, last stable fix	Published authoring data
Tale Session	Authoritative progression event, completed waypoint, chapter/block availability, replay and audit	Raw continuous sensor stream
Map Theme	Style tokens, layer policy, labels, overlays, terrain and visual treatment	Map data licensing or tile-provider credentials

5.2 Required Service Boundaries

- LandfallDefinitionService - authoring, validation, versioning, and published snapshot construction.

- LandfallMapService - map configuration, provider selection, style compilation, and attribution.
- LandfallLocationEngine - sample intake, filtering, staleness, confidence, dwell, hysteresis, and route matching.
- LandfallProgressionService - converts verified location outcomes into idempotent canonical Tale Session actions.
- LandfallProjectionService - builds Creator, Captain, Player, replay, and public-safe views.
- LandfallPrivacyPolicy - classification, coordinate precision, retention, sharing, and public-package rules.
- LandfallOfflineService - map-region packaging, asset prefetch, cache state, and reconciliation.
- LandfallTelemetryService - sanitized diagnostics without storing private paths or raw trails.

5.3 Runtime Ownership

The map renderer owns map camera transforms, symbol placement, route rendering, and map-space animation. Motion owns React layout and presence around the map. GSAP or Lanternwake scenes may coordinate narrative overlays and transitions outside the renderer, but must not fight the map engine for the same camera or marker transform. The location engine owns confidence. Tale Session services own progression. Everyone gets one steering wheel; civilization survives another day.

6. Waypoint and Route Domain

6.1 Waypoint Types

Type	Purpose
Arrival Point	Circular or shaped region representing a destination. Confirms after accuracy and dwell requirements pass.
Search Region	Broad area that tells the Player they have reached the correct search space but does not reveal an exact target.
Pass-Through Gate	A route corridor or line crossing that progresses when the Player moves through it in an allowed direction.
Dwell Zone	Requires remaining within a region for a Creator-defined period.
Sequence Waypoint	Requires earlier waypoints before it may confirm.
Optional Discovery	May reveal a side quest, log entry, image, artifact, or route without blocking the main Chronicle.
Hidden Waypoint	Exists in progression but is omitted or generalized on the Player map until a reveal condition occurs.
Moving or Temporary Waypoint	Represents a time-bounded or dynamically supplied target; restricted to approved use cases.
Indoor Region	Uses a floor-plan or site-map region plus contextual evidence; does not claim GPS-level certainty indoors.
Natural Landmark	Uses existing visual characteristics of the real place through Vision Waypoint integration; no installed marker required.

6.2 Geometry Types

- Point with radius
- Circle
- Polygon
- Multipolygon
- Route line
- Buffered route corridor
- Entrance line

- Floor-map region
- Georeferenced image overlay
- Approximate region with intentionally hidden center

6.3 Precision Profiles

Profile	Typical authoring intent	System behavior
Broad Arrival	Property, campus, park, museum, shoreline	Large region; low dwell; accepts moderate accuracy; suitable for background geofence hints.
Outdoor Waypoint	Monument, clearing, overlook, trail junction	Foreground high-accuracy samples; multiple fixes; moderate dwell; visible uncertainty.
Close Search	Bench cluster, sculpture area, courtyard feature	Requires better accuracy and longer stability; should provide a clue or contextual fallback.
Indoor Region	Room, wing, gallery, floor	Uses aligned indoor map, route context, motion continuity, and optional natural landmark evidence.
Exact Object	One exhibit or prop	Must not rely on GPS alone. Use natural visual confirmation, explicit Player observation, or optional precision provider.

6.4 Route Models

- Ordered route: waypoints must be completed in sequence.
- Flexible route: a set of waypoints may be visited in any order.
- Branching route: a choice selects one of several valid paths.
- Loop route: the Player returns to an earlier area without duplicate completion.
- Guided corridor: the system highlights a preferred path and recognizes route progress.
- Hidden route: future segments remain concealed until narrative conditions reveal them.
- Approximate route: only broad direction is shown to preserve exploration and surprise.
- Captain-directed route: an authorized Captain may reveal or replace the next destination.

7. Creator Studio Experience

7.1 Landfall Map Canvas

Creator Studio must provide a first-class Landfall editor rather than requiring coordinates to be typed into a form. The primary authoring surface is a live map canvas with the Chronicle's current style, waypoint layers, route layers, hidden-content overlays, preview controls, and device-test evidence.

- Search by address, place name, coordinates, or existing Chronicle location.
- Click or long-press to place a waypoint.
- Drag the waypoint center and handles directly on the map.
- Draw polygons, route lines, entry gates, and route corridors.
- Reorder sequence waypoints in a list or directly through route editing.
- Group waypoints by chapter, route, side quest, day, or location collection.
- Preview public, Creator, Captain, and Player visibility states.
- Preview hidden, nearby, revealed, completed, skipped, failed, and replay states.
- Switch among Chronicle map themes without losing geometry.
- Run a mobile preview using the Player's expected viewport and safe areas.

7.2 Waypoint Inspector

Setting group	Creator controls
Identity	Name, internal label, Player label, description, icon, chapter/block relationship.
Geometry	Point, radius, polygon, floor-map region, route gate, corridor width, center hiding.
Verification	Precision profile, required accuracy, sample count, dwell, hysteresis, speed limit, route match.
Presentation	Hidden until reveal, map label, marker style, pulse behavior, fog, route appearance, arrival scene.
Progression	Required, optional, sequence, completion action, side effects, retry policy, Captain override.
Guidance	Distance display, bearing display, textual clue, nearby clue, wrong-direction behavior, numeric visibility.
Privacy	Location classification, public-coordinate precision, retention, Captain visibility, export restrictions.
Fallback	Manual confirmation, clue-based completion, Captain approval, alternate waypoint, accessibility path.
Testing	Recorded fixes, observed accuracy, pass/fail history, device/browser matrix, publication warnings.

7.3 Route Authoring

- Draw a route freehand or snap it to available roads, trails, paths, or Creator-defined corridors.
- Choose whether the Player sees the full route, only the next segment, a rough bearing, or no route at all.
- Attach reveal rules to route segments.
- Define travel mode: walking, cycling, vehicle, boat, indoor, mixed, or unspecified.
- Define off-route tolerance and whether deviation produces a warning, a clue, or no response.
- Preview estimated distance and duration without treating estimates as authoritative promises.
- Detect impossible jumps, duplicate points, self-intersections, inaccessible segments, and privacy-sensitive geometry.

7.4 Creator Test Walk

Studio should support a field-testing mode that records sanitized diagnostic evidence while the Creator physically tests a route. The test walk shows accuracy, sample age, dwell state, confidence, route match, waypoint transition, and battery/network behavior. It must not become an unbounded personal location history. The Creator can retain a compact test receipt and optionally a simplified diagnostic trace for the authored route only.

7.5 Authoring Guardrails

- Warn when a waypoint radius is smaller than the selected precision profile can reasonably support.
- Block exact-object GPS confirmation without a secondary contextual or manual method.
- Warn when a route crosses restricted, dangerous, or obviously inaccessible areas when map data indicates it.
- Warn when a private real-world location would be included in a public or community release.
- Warn when background behavior is requested without a native-app requirement.
- Warn when a map style hides essential landmarks or required attribution.
- Block publication if no location-free fallback exists for a mandatory waypoint unless the Creator explicitly accepts the restricted-access classification.

8. Map Rendering and Style System

8.1 Rendering Principle

Landfall should use a vector-map rendering architecture that separates the renderer, style, data sources, and Chronicle overlays. A MapLibre-compatible approach is a strong baseline because the renderer supports interactive vector maps and the style specification defines map sources, layer order, and visual treatment through a structured style document. [R5] [R6]

Design Requirement

Map data may be ordinary. The experience must not look ordinary. Chronicle styling, authored overlays, reveal state, camera motion, route treatment, labels, fog, and marker language must transform the map into part of the story.

8.2 Built-In Style Presets

Preset	Intent
Parchment Chart	Aged paper, restrained labels, ink routes, brass markers, hand-authored overlays.
Night Harbor	Dark water and land, warm waypoint light, low-glare navigation, moonlit route treatment.
Field Journal	Paper-white foundation, pencil contours, annotation marks, documentary clarity.
Museum Blueprint	Muted blueprint or architectural treatment, floor labels, room zones, exhibit anchors.
Surveyor	High-clarity terrain, contours, paths, elevation, distance and route precision.
Minimal Light	Clean modern map with Chronicle typography and subtle thematic markers.
Minimal Dark	Low-light contemporary map for evening travel and OLED displays.
Custom Chronicle	Creator-controlled tokens and allowed layer choices built on a validated base style.

8.3 Creator Style Controls

- Land, water, road, path, building, border, label, and terrain colors.
- Label density, category visibility, hierarchy, and typography token selection.
- Road, trail, route, border, and contour line weights.
- Building and terrain emphasis.
- Hillshade, elevation, contour, satellite, or raster-overlay availability where supported.
- Chronicle waypoint icon family, marker shape, completion treatment, and hidden-state treatment.
- Fog-of-exploration and discovery masks.
- Route reveal style: ink, thread, brass line, glow, pencil, scratch, or minimal highlight.
- Camera policy: north-up, heading-up, route-up, cinematic locked, or user-controlled.
- Optional georeferenced custom image overlay, including hand-drawn maps and floor plans.

8.4 Georeferenced Custom Maps

Creators should be able to upload a custom image and align it to real geography using two or more control points, a bounding box, or corner anchors. This allows a hand-drawn treasure map, estate map, museum floor

plan, fictional nautical chart, or historical map to become the visible surface while the underlying Landfall geometry remains accurate.

The Player may experience one of three map compositions:

- Real World - vector geography with Chronicle styling.
- Authored Overlay - a custom map layered over real geography with adjustable opacity and masks.
- Authored World - the custom map is primary while hidden real geometry powers location and route logic.

8.5 Provider and Attribution Policy

Landfall must not bind the product to one commercial map vendor. The map renderer, vector or raster tile source, geocoder, routing source, elevation source, and satellite source should be separate providers. If OpenStreetMap data is used, attribution and provider policies must be honored. The OpenStreetMap Foundation explicitly distinguishes freely usable map data from its donation-funded public tile servers, which are not intended to serve unrestricted production traffic. [R7]

9. Player Experience and Voyage Chart Integration

Player Experience: One Living Chart, Three Views

The map is not embedded as a foreign product. It is skinned, narrated and synchronized with the Chronicle.

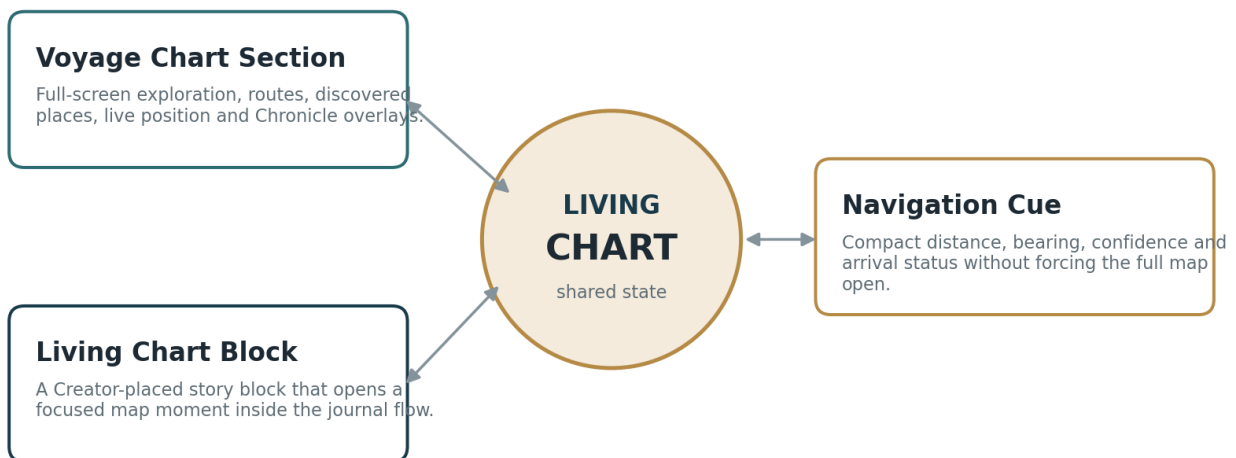


Figure 3. One Living Chart state presented through three Player views.

9.1 The Voyage Chart Is Canonical

The built-in Voyage Chart should become the primary full-screen Landfall surface. It owns the complete journey context: discovered regions, current destination, live position, route state, previous waypoints, location-triggered reveals, confidence, offline state, and replay-safe historical presentation.

9.2 Living Chart Block

The Living Chart block is a Creator-authored Story Block that presents a focused map moment inside the normal Chronicle flow. It is not a second map implementation. It opens or embeds a view into the canonical Living Chart state with a Creator-selected camera, target, controls, text, and completion policy.

- Show current location and the next destination.
- Follow the Player while they walk.
- Present a fixed cinematic view of a route or region.
- Reveal a waypoint, clue region, or new route segment.
- Require entering or dwelling in a region before completion.
- Allow the Player to open the full Voyage Chart without losing story position.
- Collapse to a readable static representation in reduced-motion or unsupported environments.

9.3 Compact Navigation Cue

A compact cue may appear in the journal header, objective card, map tab, or dedicated navigation surface. It can show direction, distance, confidence, route state, and a control to open the Voyage Chart. The cue must use plain language when numerical precision would be misleading.

Confidence state	Recommended Player language	Visual treatment
Unavailable	Location is unavailable. Continue with the clue or choose another verification path.	Static icon, explanation, fallback action.
Weak	Your position is still settling.	Large accuracy halo, no precise arrival claim.
Nearby	You have reached the surrounding area.	Region emphasis, optional nearby clue.
Likely inside	You appear to be within the waypoint.	Dwell progress, restrained confidence cue.
Confirmed	Waypoint reached.	Canonical arrival scene and progression receipt.

9.4 Player Controls

- Center on Player.
- Center on current objective.
- Fit current route.
- Toggle north-up or heading-up where supported.
- Open route and waypoint details.
- Read a text-only direction summary.
- Pause location use.
- Retry permission or location acquisition.
- Choose a location-free fallback.
- View what location information is currently being used or retained.

9.5 Arrival Experience

Arrival should feel like a Chronicle event rather than a generic pin turning green. The route may settle, the waypoint marker may resolve, the journal may reveal a note, the map may stamp the location, or an artifact may become available. The animation begins only after the canonical progression action succeeds. No triumphant brass chime before the server confirms the event, because optimism is not a transaction.

10. Chronicle Blocks and Completion Providers

10.1 Block Types

Block key	Purpose
livingChart	Full or embedded map presentation using canonical Landfall state.
waypointJourney	Guided objective centered on reaching one waypoint or region.

Block key	Purpose
routeJourney	Multi-waypoint or route-segment experience with progress and path presentation.
locationReveal	Reveals a destination, route, region, clue, map layer, or hidden overlay.
locationObservation	Asks the Player to observe something at the place and respond; useful for exact-object experiences.
locationChoice	Presents route or destination choices that become canonical branch edges.

10.2 Completion Providers

Location should also exist as a completion provider that can be attached to other block types. A narrative, artifact, riddle, photo, audio, or Captain-approval block may require Landfall evidence without forcing the Creator to insert a dedicated map block.

```
type LandfallCompletionProvider = {
  provider: "landfall";
  waypointId: string;
  requiredOutcome: "NEARBY" | "LIKELY_INSIDE" | "CONFIRMED";
  dwellSeconds?: number;
  allowCaptainOverride: boolean;
  allowPlayerFallback: boolean;
  replayPolicy: "PRESENTATION_ONLY" | "NONE";
};
```

10.3 Existing Block Integration

- A chapter release may reveal the next waypoint on the Voyage Chart.
- A riddle block may unlock its answer field only after the Player reaches the search region.
- An artifact block may grant a relic after a confirmed arrival.
- A photo or audio block may become available only at the authored location.
- A Captain approval block may combine Player location evidence with Captain confirmation.
- A Vision Waypoint block may activate only after Landfall confirms the surrounding region.
- A side quest may be discovered by passing near an optional waypoint.

11. Accuracy, Confidence, and Arrival Verification

Confidence-Aware Arrival Pipeline

No waypoint completes from one noisy dot. Evidence is filtered, bounded and reconciled before progression.



Typical outputs: Unavailable | Accuracy too low | Nearby | Likely inside | Confirmed

Fallback rule: The Player must always have a readable, fair path when location is unavailable or uncertain.

Figure 4. Confidence-aware arrival pipeline.

11.1 Location Evidence

The browser Geolocation API supplies latitude, longitude, time, and a reported accuracy radius. The W3C specification treats accuracy as part of the position data, and browser documentation describes it as a meter-based confidence radius. [R1] [R8] Landfall must treat that value as uncertain evidence rather than an exact measurement.

11.2 Sample Qualification

- Reject samples older than the waypoint or session freshness policy.
- Reject impossible coordinates and invalid accuracy values.
- Reject or heavily down-weight samples whose reported accuracy exceeds the profile threshold.
- Detect impossible speed, teleportation, or direction changes relative to recent stable state.
- Distinguish stationary, walking, driving, boating, and unknown motion where evidence permits.
- Record why evidence was rejected using safe diagnostics.

11.3 Stabilization

- Use a median, weighted median, or similarly robust filter over a bounded sample window.
- Prefer recent high-quality samples without allowing one optimistic point to dominate.
- Match samples to an authored route corridor when the Player is expected to follow a path.
- Use hysteresis so the Player does not rapidly enter and leave a waypoint at the boundary.
- Preserve a last-stable state for a short grace period through temporary signal degradation.
- Never hide the reported uncertainty from diagnostics or Creator testing.

11.4 Confirmation Rule

```

confirmed =
  freshSample
  && qualifiedAccuracy
  && insideGeometryWithUncertainty
  
```

```

&& consecutiveEvidence >= requiredSamples
&& dwellElapsed >= requiredDwell
&& routeOrMotionConstraintsSatisfied
&& noTerminalPermissionOrIntegrityFailure;

```

11.5 Boundary Mathematics

The verifier should account for both the waypoint geometry and the sample uncertainty. A Player whose estimated point is two meters inside a ten-meter waypoint but whose reported accuracy is sixty meters should not be treated as definitely present. Conversely, a Player with excellent accuracy near the boundary may satisfy a soft or dwell-based rule. The policy must be explicit per precision profile.

11.6 Background Geofences

Background geofences should be broad arrival hints, not exact progression gates. Android guidance recommends geofence radii around 100 to 150 meters for best results, reflecting the realities of background location and device power management. [R4] A native Landfall app may use a broad geofence to wake the experience, then perform foreground confidence verification when the Player opens the Chronicle.

11.7 Integrity and Spoofing

Consumer location can be spoofed or manipulated. Landfall may detect obvious impossible movement and provide integrity metadata, but it must not claim security-grade proof. For personal Chronicles, narrative fairness matters more than adversarial enforcement. High-stakes or competitive uses should require another authoritative method or Captain confirmation.

12. Small-Area and Indoor Experiences Without Physical Setup

12.1 Governing Approach

Indoor and exact-object experiences must be designed around layered confidence rather than a fantasy that satellites can distinguish adjacent display cases. Landfall first confirms the broad place, then narrows using the authored map, route, motion continuity, and natural features already present in the environment.

12.2 Natural Landmark Verification

A Creator may capture reference images of an existing doorway, mural, exhibit, statue, sign, architectural detail, or landscape feature. The Player's camera can compare the current view through the existing Vision Waypoint architecture. Nothing is installed at the site. The place itself is the marker.

- Reference sets should include several angles, distances, lighting conditions, and negative examples.
- Landfall supplies the broad region so the vision system searches only where the landmark is plausible.
- Vision results remain confidence-based and may require several frames.
- Raw camera frames should be retained only under an explicit privacy policy.
- A readable manual observation or Captain fallback must exist for critical progression.

12.3 Floor Maps and Site Maps

Creators may upload a floor map or site diagram and align it to real geography. Rooms, wings, exhibits, corridors, stairs, and entrances can be drawn as regions or paths. The map may be fully private to the Chronicle.

Public community releases must not reveal sensitive private layouts unless the Creator has explicit rights and the classification permits it.

12.4 Contextual Positioning Without Installed Infrastructure

Evidence	Use	Limitation
Outdoor or entry GPS	Confirms property or building vicinity.	Poor indoors; multipath near buildings.
Route continuity	Estimates progress along a Creator-authored corridor.	Drifts when the app is suspended or motion is ambiguous.
Device heading and motion	Supports directional guidance and short continuity.	Magnetic interference and sensor drift.
Barometric/elevation change	Suggests floor or stair movement.	Device variability and weather effects.
Natural landmark vision	Confirms existing scene or exhibit.	Lighting, occlusion, crowding, and model confidence.
Player observation	Asks for an answer visible only at the target.	Not automatic; intentionally relies on human understanding.
Captain confirmation	Authorized human fallback.	Requires Captain availability.

12.5 Museum Experience Pattern

1. A broad waypoint confirms arrival at the museum property.
2. A floor-map route guides the Player to the correct wing or gallery.
3. Motion continuity and route matching maintain approximate progress indoors.
4. A natural landmark or observation clue confirms the target area.
5. The exact exhibit is identified through what the Player sees, not through an imaginary two-centimeter GPS fix.
6. The Chronicle reveals the narrative, artifact, or next direction through canonical progression.

13. Permissions, Privacy, and Safety

13.1 Permission UX

Web geolocation requires explicit permission and secure contexts, and it may be restricted by the site's Permissions Policy. [R2] [R9] Landfall must ask only when the Player enters a feature that needs location, after explaining the benefit and the available fallback.

- Do not request location on the public landing page or during unrelated account creation.
- Explain whether the Chronicle needs foreground location, background location, camera, motion, or notifications.
- Explain what is processed locally, what is stored, who can see it, and how to pause it.
- Provide a visible location-active state and a one-action pause control.
- Request background permission only when a native Tier III feature clearly needs it.
- Do not punish a Player for choosing approximate location; adapt the guidance and offer alternatives.

13.2 Location Classification

Class	Meaning	Public behavior
FICTIONAL	Not tied to real geography.	No coordinate privacy concern.
GENERIC	Broad non-sensitive place type or generalized region.	May be public in generalized form.
PUBLIC_REAL_WORLD	A public place intentionally identified exactly.	Exact location may be published when rights and safety permit.
APPROXIMATE_REAL_WORLD	Real place with deliberately reduced	Public projections generalize coordinates

Class	Meaning	Public behavior
	precision.	and route detail.
PRIVATE_REAL_WORLD	Home, private venue, sensitive layout, workplace, school, or private route.	Exact coordinates and geometry excluded from public/community projection.

13.3 Data Minimization

The normal durable record should contain waypoint identity, verification time, verification method, confidence class, and canonical progression event. It should not contain every raw latitude and longitude sample.

```
type LandfallCompletionEvidence = {
  waypointId: string;
  verifiedAt: string;
  outcome: "NEARBY" | "LIKELY_INSIDE" | "CONFIRMED";
  method: "GEOLOCATION" | "CONTEXTUAL" | "VISION" | "CAPTAIN" | "MANUAL";
  confidenceClass: "LOW" | "MEDIUM" | "HIGH";
  diagnosticReceiptId?: string;
};
```

13.4 Live Sharing

Captain live-location viewing must be off by default and must require explicit Player consent for a defined duration and purpose. Ordinary Captain views should show waypoint progress, last known confidence, and connection state rather than a continuously moving dot. A Creator publishing a Chronicle does not gain a surveillance feed merely because they placed a waypoint.

13.5 Safety Controls

- Allow Creators to mark waypoints as daylight-only, weather-sensitive, accessibility-limited, or Captain-supervised.
- Allow the platform to suppress or warn about unsafe routing contexts.
- Do not encourage crossing roads, rail lines, restricted property, water, cliffs, or private boundaries based only on a map line.
- Never require interacting with the map while driving or operating a vessel.
- Allow immediate skip, reroute, or Captain override in unsafe conditions.
- Retain a clear distinction between narrative guidance and professional navigation or emergency advice.

14. Offline and Weak-Network Operation

14.1 Web Foundation

- Cache the Chronicle shell, already-revealed content, selected map style, waypoint definitions, route geometry, and bounded map regions before play.
- Display which route areas and media are available offline.
- Continue local foreground location and confidence calculation when network access is lost.
- Queue completion evidence locally with an explicit “not yet synchronized” state.
- Never display server-confirmed success until the queued progression action is acknowledged.
- Reconcile duplicate or stale evidence idempotently when connectivity returns.

14.2 Native Foundation

Tier III should support downloadable offline map packages, background geofence registration, native cache management, and robust return from notifications. The package must contain only the map data, styles, routes, and Chronicle assets authorized for that Player and published version.

14.3 Offline Fairness

A Player should not lose access to an already-revealed chapter or route because the network disappeared. At the same time, the app must clearly distinguish local evidence, queued progression, and server-confirmed progression. The interface should not stage a finale because a request is sitting in an outbox feeling optimistic.

15. Data Model and Service Contracts

15.1 Suggested Core Types

```
type LandfallMap = {
  id: string;
  taleId: string;
  name: string;
  coordinateSystem: "WGS84" | "CUSTOM_GEOREFERENCED";
  baseStyleId: string;
  customStyleTokens: LandfallStyleTokens;
  dataProviderPolicy: LandfallProviderPolicy;
  defaultCamera: LandfallCamera;
  offlinePolicy: LandfallOfflinePolicy;
  privacyClassification: LandfallLocationClassification;
};

type LandfallWaypoint = {
  id: string;
  mapId: string;
  geometry: GeoJSON.Geometry;
  waypointType: LandfallWaypointType;
  precisionProfile: LandfallPrecisionProfile;
  requiredAccuracyMeters?: number;
  requiredSamples: number;
  dwellSeconds: number;
  enterHysteresisMeters: number;
  exitHysteresisMeters: number;
  sequencePolicy: LandfallSequencePolicy;
  visibilityPolicy: LandfallVisibilityPolicy;
  completionPolicy: LandfallCompletionPolicy;
  fallbackPolicy: LandfallFallbackPolicy;
  privacyClassification: LandfallLocationClassification;
};
```

15.2 Runtime Types

```
type LandfallRuntimeState = {
  sessionId: string;
  activeMapId?: string;
  activeWaypointId?: string;
  activeRouteId?: string;
  permissionState: "PROMPT" | "GRANTED" | "DENIED" | "UNAVAILABLE";
  trackingState: "IDLE" | "ACQUIRING" | "TRACKING" | "PAUSED" | "FAILED";
  confidence: LandfallConfidence;
  lastStableFix?: SanitizedLocationFix;
  routeProgress?: LandfallRouteProgress;
```

```

dwellProgress?: LandfallDwellProgress;
offlineState: "ONLINE" | "OFFLINE_READY" | "OFFLINE_PARTIAL" | "OFFLINE_UNAVAILABLE";
};

```

15.3 API and Action Surface

Action	Purpose
GET map projection	Return Player-safe current map, route, waypoint, style, and offline metadata.
POST location evidence	Submit bounded sanitized evidence or a verified client receipt.
POST waypoint completion	Request canonical progression using idempotency and expected session state.
POST Captain override	Authorized completion, skip, replacement, or correction.
POST Creator test receipt	Store sanitized test-walk evidence for authoring validation.
GET offline package manifest	Return authorized assets and map regions for the pinned published version.
POST privacy consent	Record time-bounded consent for optional live sharing or diagnostic retention.

15.4 Published Snapshot Requirements

- Every Landfall definition in a live Tale Session must come from the pinned immutable PublishedTaleVersion.
- Runtime changes do not mutate the published geometry or style.
- Creator edits create a new draft and later a new published version.
- Active sessions remain pinned unless an authorized migration process exists.
- Public listing metadata must not contain private geometry, hidden destinations, exact private coordinates, or Creator-only test traces.

16. Progression, Events, Replay, and Captain Controls

16.1 Event Model

- LANDFALL_TRACKING_STARTED
- LANDFALL_PERMISSION_CHANGED
- LANDFALL_WAYPOINT_NEARBY
- LANDFALL_WAYPOINT_ENTERED
- LANDFALL_WAYPOINT_DWELL_STARTED
- LANDFALL_WAYPOINT_CONFIRMED
- LANDFALL_WAYPOINT_FAILED
- LANDFALL_ROUTE_DEVIATED
- LANDFALL_ROUTE_RESUMED
- LANDFALL_OFFLINE_QUEUED
- LANDFALL_EVIDENCE_RECONCILED
- LANDFALL_CAPTAIN_OVERRIDE
- LANDFALL_FALLBACK_USED

Not every diagnostic transition belongs in the authoritative Tale Session event history. Nearby, accuracy, and dwell updates may remain ephemeral runtime state. Durable events should record progression, meaningful Player-visible changes, consent, overrides, and reconciliation outcomes.

16.2 Replay

Replay is presentation-only. Replaying a waypoint arrival may re-show the map stamp, route reveal, chapter message, or artifact ceremony using persisted safe event data. It must never re-request location, re-complete the waypoint, duplicate an artifact, or create another viewed record.

16.3 Captain Controls

- View current waypoint status, confidence class, permission state, and last synchronization time.
- Reveal a hidden waypoint or route segment.
- Replace or bypass a waypoint when conditions change.
- Approve a manual fallback.
- Confirm arrival when the Player is visibly present but sensors fail.
- Pause or resume location progression.
- Never view exact live Player position unless the Player has granted explicit time-bounded consent.

16.4 Idempotency and Ordering

Each completion request must include the Tale Session, waypoint, published version, expected sequence, evidence identity, and idempotency key. Duplicate submissions, offline retries, repeated foreground callbacks, and reconnect reconciliation must produce one canonical completion.

17. Accessibility and Inclusive Wayfinding

- Provide a text-only direction summary: distance band, direction, route step, waypoint name, and confidence.
- Announce permission, acquisition, nearby, dwell, confirmed, offline, deviation, and fallback states through appropriate live regions.
- Provide keyboard access to map controls, waypoint lists, route details, style toggles, and fallback actions.
- Provide non-map list and text views for Players who cannot use a spatial map effectively.
- Do not require compass heading from Players whose devices or movement conditions make it unreliable.
- Do not require rapid walking, precise hand motion, camera use, sound, vibration, color discrimination, or small-target interaction without an alternative.
- Respect reduced motion by replacing map fly-throughs, marker travel, and route drawing with immediate but staged state changes.
- Ensure high-contrast, forced-colors, text zoom, and screen-reader behavior for map-adjacent controls.
- Allow Creators to declare route accessibility, stairs, terrain, distance, seating, lighting, and sensory considerations.

17.1 Accessible Map Semantics

The rendered map canvas itself may not expose enough semantics. Landfall must maintain an accessible parallel structure containing current position status, active waypoint, waypoint list, route steps, distance bands, arrival state, and available actions. Hidden or unrevealed locations must remain absent from that structure.

18. Animation and Experience Language

Landfall must integrate with Project Lanternwake rather than creating a local collection of marker animations. The map renderer owns map-space transitions. Lanternwake coordinates narrative arrival scenes, map-to-journal handoffs, route reveals, waypoint stamps, and artifact or chapter ceremonies.

18.1 Scene Family

- landfall-map-arrival
- landfall-location-acquiring
- landfall-waypoint-nearby
- landfall-waypoint-confirmed
- landfall-route-reveal
- landfall-route-deviation
- landfall-offline-transition
- landfall-reconnect-reconciliation
- landfall-floor-map-transition
- landfall-natural-landmark-confirmed

18.2 Motion Rules

- No map fly-to may strand the Player away from their current objective without a clear return control.
- Camera travel must be interruptible and must settle to a deterministic state.
- Route drawing begins after the route is actually revealed in canonical state.
- Arrival animation begins after authoritative completion succeeds.
- Continuous location marker movement must be smoothed but must not imply greater accuracy than the evidence.
- Reduced motion preserves camera context through cuts, highlights, and text rather than long travel.
- Background tracking must not continuously animate hidden or offscreen map surfaces.

18.3 Experience Language

Operational language must remain clear. The system may say “The chart is finding your position,” “You have reached the surrounding area,” or “The waypoint is confirmed.” It should not bury permission errors or accuracy problems under theatrical wording. Security and navigation are poor places for a riddle about why the browser denied access.

19. Performance and Operational Architecture

19.1 Map Performance

- Load the map renderer only on routes or blocks that require it.
- Keep Chronicle overlays in bounded layers and avoid unbounded DOM markers.
- Cluster or simplify large waypoint sets.
- Use vector styles and data-driven layers where practical.
- Pause continuous rendering and location updates when the map is hidden and the active feature does not require tracking.
- Cap terrain, hillshade, blur, glow, and custom WebGL effects on constrained devices.
- Prefetch only the map region required by the current route or published offline package.
- Keep style changes from destroying and recreating the entire Landfall runtime state.

19.2 Provider Abstractions

Provider	Examples of responsibility
MapRendererProvider	Web or native map renderer, style application, layers, camera, hit testing.
MapDataProvider	Vector/raster tiles, attribution, coverage, cache rules.

Provider	Examples of responsibility
GeocodingProvider	Place search, address lookup, reverse lookup, rate and privacy policy.
RoutingProvider	Optional route suggestions, travel modes, path metadata.
ElevationProvider	Terrain, contour, elevation profile, floor or altitude hints.
LocationProvider	Browser geolocation, native Core Location, Android location services.
ContextProvider	Motion, heading, barometer, natural landmark or vision evidence.

19.3 Operational Truth

The application must report whether a provider is configured, degraded, unavailable, rate-limited, or unsupported. It may not label missing elevation, routing, map data, or scanner behavior as complete merely because the interface can render an empty layer. Software has enough decorative green checkmarks already.

20. Failure and Fallback Behavior

Failure	Required behavior
Permission denied	Explain how to enable permission, keep the Chronicle usable, and show Creator-defined fallback.
Approximate location only	Increase uncertainty, use broader states, and avoid impossible precision requirements.
Position unavailable	Retry with bounded backoff; retain last-stable context briefly; offer fallback.
Accuracy too low	Show settling state and accuracy halo; do not complete; allow nearby clue or alternate verification.
Map data failed	Use cached/custom map, route list, text directions, or static map fallback.
Geocoder failed	Authoring can continue through coordinates, map click, saved places, or retry.
Offline	Continue downloaded route and local confidence; queue progression evidence visibly.
Route mismatch	Explain deviation without panic; offer return, reroute, or Creator-authored alternate.
Indoor uncertainty	Shift to floor-map context, natural landmark, observation, or manual/Captain fallback.
App suspended	Resume from safe state; reject stale evidence; reconcile rather than replaying success.
Server rejected completion	Reverse pending presentation, preserve evidence for diagnostics, and keep the waypoint retryable.

20.1 Fallback Hierarchy

7. Retry the same in-system method when the failure appears temporary.
8. Use a broader confidence state or nearby clue when exact confirmation is unnecessary.
9. Use natural landmark or observation confirmation when the place itself provides evidence.
10. Use a Creator-authored manual path or Captain confirmation.
11. Use optional QR, NFC, beacon, or UWB providers only when the Creator intentionally chose them and the experience supports them.

Fairness Rule

A mandatory Chronicle may require visiting a place, but it must not silently require one brand of phone, one permission mode, one sensor, or a perfect sky view unless that limitation is declared before the Player begins.

21. Security and Integrity

- Validate all Creator geometry, style, provider, and progression input with strict schemas.
- Prevent arbitrary map-style URLs, scripts, unsafe custom WebGL code, or untrusted executable content.
- Keep provider credentials server-side or use restricted public tokens with domain, package, and quota controls.
- Rate-limit geocoding, routing, evidence submission, and Captain override routes.
- Use CSRF protection for state-changing cookie-authenticated actions.
- Authorize every private map, waypoint, route, test receipt, and offline package through canonical account and Tale Session policy.
- Never use public listing slugs as sufficient authorization for private geometry.
- Prevent exact private coordinates from entering logs, analytics, Open Graph metadata, screenshots, or community search documents.
- Detect replayed or duplicated evidence through idempotency and session sequence.
- Treat location as evidence, not trusted identity or payment authorization.

21.1 Diagnostic Privacy

Diagnostics may record waypoint ID, sample age, accuracy band, distance band, confidence transition, route-match result, error code, and timing. Default diagnostics must not record full raw coordinates. Exact coordinates may appear only in an explicitly authorized Creator test receipt with bounded retention and access.

22. Testing and Acceptance Matrix

22.1 Core Unit Tests

- Distance, bearing, polygon inclusion, route corridor, and boundary uncertainty.
- Stale, invalid, impossible-speed, weak-accuracy, and duplicate samples.
- Median filtering, route matching, hysteresis, dwell, and last-stable grace.
- Every precision profile and confidence transition.
- Idempotent waypoint completion and offline retry.
- Privacy projection and coordinate generalization.
- Map style schema, provider policy, attribution, and unsafe URL rejection.
- Georeferenced overlay transformation and bounded image geometry.

22.2 Creator Tests

- Place, move, resize, delete, duplicate, reorder, and hide waypoints.
- Draw, edit, split, branch, and validate routes.
- Align a custom map or floor plan.
- Change map style without losing geometry.
- Receive warnings for unrealistic precision and unsafe privacy classification.
- Run a test walk and inspect the sanitized receipt.
- Publish an immutable version and prove active sessions remain pinned.

22.3 Player Browser Matrix

Dimension	Required cases
Permission	Prompt, granted precise, granted approximate, denied, revoked, unavailable.
Environment	Open sky, trees, urban canyon, beside building, indoors, moving vehicle, stationary.
Network	Online, slow, offline-ready, offline-partial, reconnect, duplicate response.
Map	Warm cache, cold cache, provider failure, style failure, custom overlay, floor map.
Progression	Nearby, likely inside, confirmed, dwell interrupted, route deviation, Captain override, replay.
Input	Touch, mouse, keyboard, screen reader, reduced motion, 200% zoom.
Viewport	Desktop, tablet, portrait phone, landscape phone, narrow-height desktop.

22.4 Native Matrix

- Foreground, background, screen locked, app terminated, notification return, permission downgrade, battery saver, low-power mode.
- Geofence entry, delayed entry, false exit, broad wake plus foreground confirmation.
- Offline map package download, expiration, update, partial failure, and cleanup.
- Location updates after app upgrade and published-version change.

22.5 Privacy and Security Tests

- Private coordinates excluded from public projections, community releases, logs, analytics, screenshots, and metadata.
- Non-owner cannot view Creator test traces or private routes.
- Captain cannot view exact location without active consent.
- Revoked consent stops new sharing immediately.
- Provider keys and restricted tokens are not exposed improperly.
- Location evidence cannot mutate another session or waypoint.

23. Implementation Phases

23.1 Phase 1: Lay the Bearings

Foundation, canonical domain, map provider abstraction, privacy model, type system, map renderer, basic live location, confidence engine, and isolated validation.

- Landfall design record and threat model.
- Map, waypoint, route, style, privacy, and runtime types.
- MapLibre-compatible web renderer and provider seam.
- Foreground geolocation and confidence pipeline.
- Basic Voyage Chart live-position layer.
- No public publishing and no background tracking.

23.2 Phase 2: Bring the World Aboard

Creator Studio authoring, Chronicle blocks, complete Voyage Chart integration, route and style editing, test walks, progression events, Captain controls, replay, and offline web foundation.

- Live map authoring canvas.
- Waypoint and route inspectors.
- Living Chart, waypoint journey, route journey, and completion provider.
- Custom map overlays and built-in themes.
- Player arrival and route scenes through Lanternwake.
- Field test receipts and authoring warnings.

23.3 Phase 3: Read the Ground

Contextual precision, floor maps, route matching, natural landmark Vision Waypoints, indoor guidance, observation-based exact-target patterns, and richer confidence fusion without required physical infrastructure.

- Floor and site map alignment.
- Natural landmark reference workflows.
- Indoor region and route continuity.
- Motion, heading, elevation, and context fusion.
- Museum and compact-area test suite.
- No claim of impossible GPS precision.

23.4 Phase 4: Hold the Bearing

Native companion app integration, background geofencing, offline map packages, notifications, native sensor fusion, advanced optional providers, production infrastructure, privacy review, performance, and launch readiness.

- iOS and Android native location provider.
- Background and notification workflows.
- Offline region packaging and cache operations.
- Power and permission governance.
- Optional UWB, beacon, QR, or NFC providers behind explicit capability flags.
- Production monitoring, incident procedures, and full acceptance.

24. Final Acceptance Criteria

Gate	Required result
Canonical state	Voyage Chart, embedded blocks, compact cues, Creator previews, and Captain views use one Landfall domain.
Zero infrastructure	A complete outdoor and contextual Chronicle can be authored and played without installed markers or hardware.
Native presentation	The Player map is Chronicle-styled and integrated, not a pasted third-party interface.
Creator authoring	Creators can visually place waypoints, draw routes, create regions, style maps, align custom maps, and test behavior.
Confidence truth	No mandatory waypoint completes from one weak, stale, or unqualified sample.
Indoor honesty	Exact indoor/object use cases have contextual, natural landmark, observation, manual, or Captain paths.
Privacy	Raw trails are not stored by default; private coordinates are absent from public projection and ordinary analytics.

Gate	Required result
Permissions	Location is requested contextually, can be paused, and has a usable fallback.
Progression	Every completion is canonical, idempotent, version-pinned, replay-safe, and auditable.
Offline	Downloaded content remains readable; local evidence is distinguished from server-confirmed progress.
Accessibility	Map and navigation state have text, keyboard, screen-reader, reduced-motion, and location-free alternatives.
Provider independence	Renderer, tiles, geocoding, routing, elevation, and location services are replaceable.
Operations	Attribution, cache policy, quotas, provider health, and failure states are truthful.
Testing	Required browser, device, environment, permission, network, privacy, and progression matrices pass.

Completion Rule

Project Landfall is not complete because a dot can move on a map. It is complete when Creators can author fair journeys, Players can understand and trust the guidance, progression remains correct, private places remain private, and the experience survives the environment behaving like the environment.

25. Recommended Technical Baseline

25.1 Web

Area	Recommended baseline
Map renderer	MapLibre GL JS or an equivalent renderer compatible with a structured vector style specification.
Map styles	Validated style documents plus Chronicle style tokens and allowed layer overrides.
Location	W3C Geolocation API through a centralized provider using watched foreground updates.
Geometry	GeoJSON-compatible geometry with a vetted geospatial library for distance, polygon, buffering, and route operations.
State	Canonical server projections plus bounded client runtime state; no independent map-component truth.
Offline	Service worker, versioned map-region manifest, authorized asset cache, and explicit synchronization outbox.

25.2 Native

Area	Recommended baseline
Map renderer	MapLibre Native or equivalent native renderer sharing style and data contracts with web.
iOS location	Core Location provider with explicit foreground/background modes and permission policy.
Android location	Platform location and geofencing provider with broad geofences plus foreground verification.
Offline	Native region downloads, encrypted authorized manifests, version-pinned cache, bounded retention.
Sensors	Explicit providers for heading, motion, barometer, and optional advanced capabilities.

25.3 Map Data

Select a production tile and geocoding strategy only after reviewing coverage, cost, cache rights, attribution, offline terms, privacy, rate limits, and self-hosting requirements. OpenStreetMap data can underpin the system, but public OSM tile infrastructure must not be treated as a free production CDN. [R7]

25.4 Decision Recommendation

Recommended Starting Architecture

Use a MapLibre-compatible renderer, a provider-agnostic map data interface, the existing Voyage Chart as the canonical Player surface, and a web-first Tier I implementation. Build Tier II through route context, aligned custom maps, and natural landmarks before adding required physical-world markers. Reserve native background and advanced precision for Tier III.

26. Glossary

Term	Definition
Living Chart	The canonical Landfall map state and experience shared by Voyage Chart, blocks, cues, previews, and replay.
Waypoint	A point, region, corridor, gate, or landmark tied to guidance or progression.
Search Region	A broad destination that confirms the Player is in the correct area without identifying an exact object.
Precision Profile	A validated policy defining accuracy, sample, dwell, context, and fallback expectations.
Confidence	A typed interpretation of evidence: unavailable, weak, nearby, likely inside, or confirmed.
Dwell	The duration the Player must remain within qualified waypoint evidence before confirmation.
Hysteresis	Different entry and exit tolerances that prevent state flapping near a boundary.
Route Matching	Comparing location evidence to an authored path or corridor.
Natural Landmark	An existing real-world scene or object used for contextual or vision confirmation without installed infrastructure.
Georeferenced Overlay	An authored image aligned to real coordinates, such as a floor plan or treasure map.
Background Geofence	An OS-managed broad region used to wake or notify a native app, not to prove exact arrival.
Location Evidence	Bounded facts used to request canonical progression, distinct from continuous raw tracking.

27. References

[R1] W3C, Geolocation specification. Provides position, accuracy, time, one-shot updates, and watched updates. [Official source](#)

[R2] MDN, Geolocation API. Documents secure-context, permission, and Permissions Policy considerations. [Official source](#)

[R3] Apple Developer Documentation, Core Location. Documents standard/significant updates, region monitoring, and related location services. [Official source](#)

[R4] Android Developers, Create and monitor geofences. Includes background geofence guidance and recommended broad radii. [Official source](#)

[R5] MapLibre GL JS documentation. Describes the WebGL vector-map renderer and map controls. [Official source](#)

[R6] MapLibre Style Specification. Defines map sources, layers, order, and visual styling through structured style documents. [Official source](#)

[R7] OpenStreetMap Foundation Operations Working Group, Tile Usage Policy. Distinguishes open map data from limited public tile infrastructure. [Official source](#)

[R8] MDN, GeolocationCoordinates accuracy. Describes meter-based reported accuracy at a 95% confidence level. [Official source](#)

[R9] MDN, Permissions-Policy geolocation directive. Describes origin control for geolocation access. [Official source](#)

Closing Declaration

Project Landfall governs how Chronicles meet the physical world. It must make real places feel like authored story spaces while remaining honest about uncertainty, protective of privacy, accessible to Players, practical for Creators, and resilient when technology behaves like technology instead of a promotional animation.

Final Identity

PROJECT LANDFALL • THE LIVING WORLD NAVIGATION SYSTEM